BANKWISE ATM - POS - CARD STATISTICS (2022-2025)

1. Project Overview and Objective

This project involves cleaning, transforming, and analyzing usage of CREDIT CARD, DEBIT CARD and QR Code in various terminal from different bank accounts using Excel and building an interactive Power BI dashboard to derive meaningful business insights.

The main objective is to centralize monthly data, ATM deployment Trends, Card usage and its growth, POS terminal penetration and to classify segment banks into categories.

2. Data Sources

Source : India data port/Economy/Reserve Bank of India/2022-2025.

Domain : Finance/Banking .

3. Problem Statement

To analyze

- Category and Name of the Bank.
- Usage of ATM's in on Onsite and offsite.
- Usage of Micro ATM's
- Payments done at Point of Sale.
- Transactions done by QR Code and UPI.
- Total outstanding Credit and Debit cards.
- Details of Usage of Credit and Debit cards in POS and Online.
- Details of Usage og Credit and Debit cards for Cash Withdrawl.

Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Id	Number	Identification number
Date	Date Format	Date for the transaction
Bank_Category	TEXT	Category of the Bank
Bank_Name	TEXT	Name of the bank in each category

ATM's Onsite	Number	Usage of ATM located near Bank Premises
ATM's Offsite	Number	Usage of ATM located in public places and other terminals
POS	Number	Point of Sale
Micro_ ATM's	Number	doorstep banking services in remote or underserved areas.
Bharath_QR	Number	unified, interoperable QR code payment system
Upi_QR	Number	Unified Payments Interface (UPI) system in India.Cashless payments by Scanner.
Credit_Cards	Number	Outstanding Credit Cards
Debit Cards	Number	Outstanding Debit Cards
No of CC at POS	Number	Total Credit Cards at Point of Sale
Value of CC at POS	Currency	Total value spent by Credit Card holders on point of sale
No of CC Online Transaction	Number	Total number of Credit Cards used for Online Transactions.
CC Online Transaction value	Currency	Total value spent by Credit Card holders in Online Transactions.
No of other CC Transactions	Number	Total number of Credit Cards used for other Transactions.
Value of other CC Transactions	Currency	Total value spent by credit card holders for other transactions.
No of Credit CC Withdrawl	Number	Number of credit card used for cash withdrawl.
Value of CC_Cash Withdrawl	Currency	Total value spent by credit card holders for cash withdrawl
No of DC at POS	Number	Number of Debit card at Point of sale

Value of DC at POS	Currency	Total value spent by Debit card holders at Point of sale
No of DC Online Transaction	Number	Number of Online Transactions using debit card
Value of DC Online Transaction	Currency	Total value spent in Online transactions using debit card
No of other DC Transactions	Number	Number of other Debit card Transactions
Value of other DC Transactions	Currency	Total amount of other debit card transactions
No of DC _Cash Withdrawl_ATM	Number	Number of debit card used for cash withdrawl
Value_ DC _Cash Withdrawl_ATM	Currency	Total value of transactions used by debit card for cash withdrawl
No of DC _Cash Withdrawl_POS	Number	Number of debit card used at POS terminal for cash withdrawl
Value_DC _Cash Withdrawl_POS	Currency	Total value of Debit card used at POS
CC_Transaction Total Value	Currency	Total transaction value of credit card used in all terminal
DC_Transaction Total value	Currency	Total transaction value of debit card used in all terminal
Comparision of CC and DC Total value	Currency	Comparing Debit and credit card Total value

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning & Transformation :

- No duplicates found.
- Standardized formats as Currency, Date, Number and Text.
- Created calculated columns Total value of Credit cards and Debit cards using SUM Function.
- Compared total value of credit and debit cards using IF function.

- Performed conditional formating by Highlighting cell rules.

Filtering & Sorting: Used filter for Category of Bank and seperated into multiple sheets

PIVOT TABLE 1: ANLYSIS OF ATM, POS AND QR CODE

Bank_Category	Sum of ATM's Onsite	Sum of ATM's Offsite	Sum of POS	Sum of Bharath_QR	Sum of Upi_QR
Foreign Banks	26949	28183	1896525	43	229917
Payment Banks	24	1075	14840776	0	906341846
Private Sector Banks	1885127	1419845	293704894	158458082	1619293395 7
Public Sector Banks	3368305	2460661	66563002	87464994	647463332
Small Finance Banks	124119	1084	647194	2087990	71747061
Grand Total	**5404524**	**3910848**	**377652391**	**248011109**	**17818716113**

PIVOT TABLE 2: COMPARISION OF PUBLIC AND PRIVATE SECTOR BANKS USING CREDIT AND DEBIT CARDS

	Bank_Category		
Values	Private Sector Banks	Public Sector Banks	Grand Total
Sum of Value of CC at POS	18271263774	4851172343	23122436117
Sum of CC Online Transaction value	30909305666	8162992388	39072298054
Sum of Value of other CC Transactions	120332.78	3620.13	123952.91
Sum of Value of CC_Cash Withdrawl	106187901.9	50308524.03	156496426
Sum of Value of DC at POS	₹ 5,896,621,081.64	₹ 7,695,096,186.92	₹ 13,591,717,268.56
Sum of Value of DC Online Transaction	₹ 4,133,866,244.42	₹ 2,612,731,650.46	₹ 6,746,597,894.88
Sum of Value of other DC Transactions	₹ 103,053,811.71	₹ 10,108,963.26	₹ 113,162,774.97
Sum of Value_ DC _Cash Withdrawl_ATM	₹ 34,775,579,453.12	₹ 75,376,920,618.43	₹ 110,152,500,071.55
Sum of Value_DC _Cash Withdrawl_POS	₹ 1,651,688.61	₹ 3,059,152.45	₹ 4,710,841.06

PIVOT TABLE 3: COMPARISION OF ATM ONSITE,OFFSITE AND MICRO ATM

FOR FINANCIAL YEAR(2022-2023)

Used Filter for selecting year and date

Date	(Multiple Items)		
Bank_Category	**Sum of ATM's Onsite**	**Sum of ATM's Offsite**	**Sum of Micro_ ATM's**
Foreign Banks	8265	12283	3462189.00
Payment Banks	12	764	8140697.00
Private Sector Banks	476495	431706	1732671.00
Public Sector Banks	934103	717570	1900446.00
Small Finance Banks	30648	280	35021.00
Grand Total	**1449523**	**1162603**	**15271024.00**

FOR FINANCIAL YEAR (2023-2024)

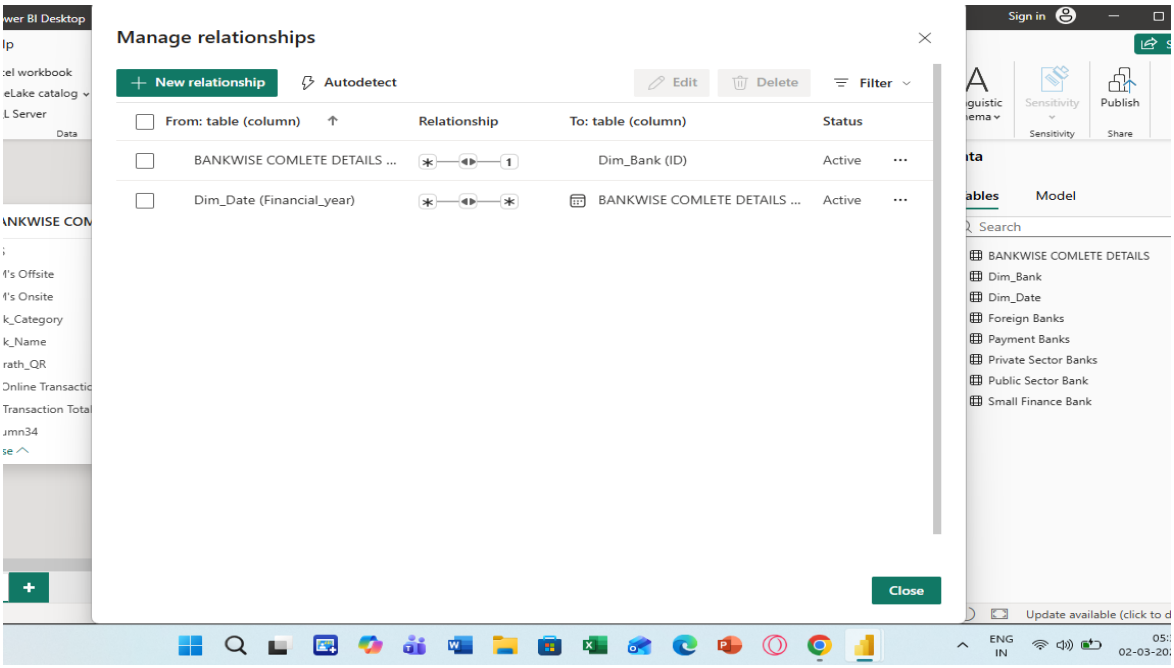
Date	(Multiple Items)		
Bank_Category	**Sum of ATM's Onsite**	**Sum of ATM's Offsite**	**Sum of Micro_ ATM's**
Foreign Banks	7296	7091	563381.00
Payment Banks	11	241	14109263.00
Private Sector Banks	523829	418561	2164943.00
Public Sector Banks	937472	701924	2069179.00
Small Finance Banks	35336	288	58151.00
Grand Total	**1503944**	**1128105**	**18964917.00**

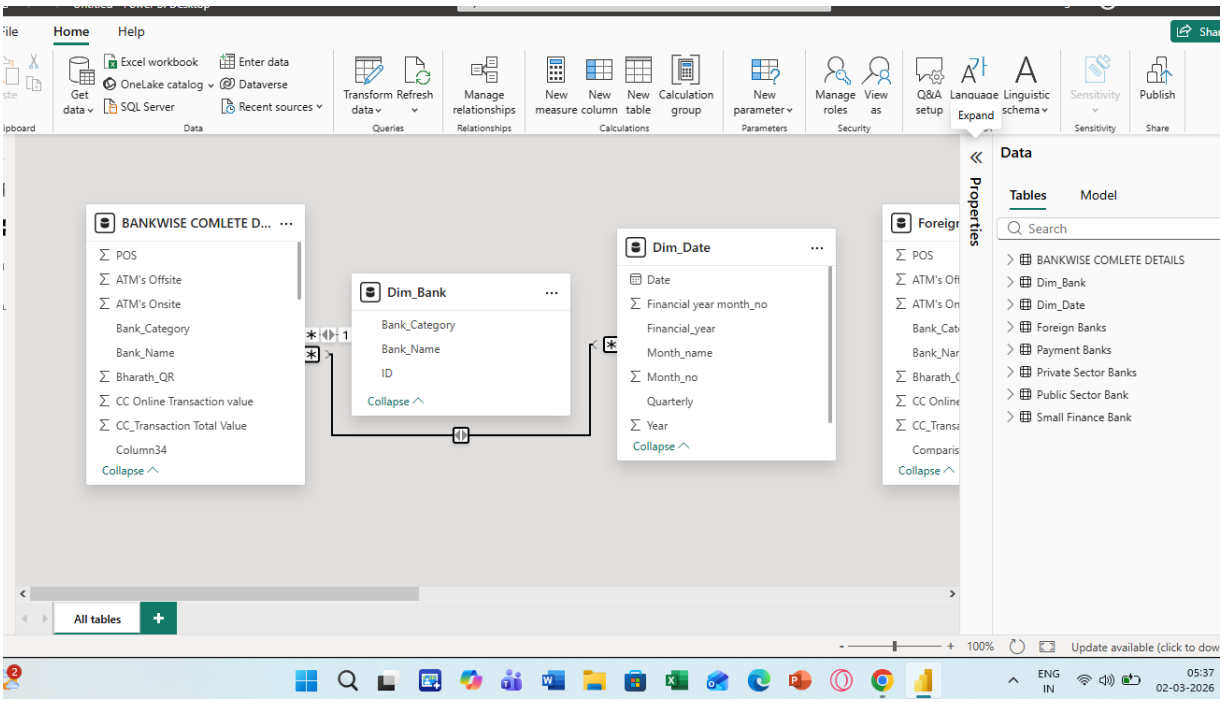
FOR FINANCIAL YEAR(2024-2025)

Date	(Multiple Items)		
Bank_Category	**Sum of ATM's Onsite**	**Sum of ATM's Offsite**	**Sum of Micro_ ATM's**
Foreign Banks	7217	5090	7359.00
Payment Banks	0	0	13508881.00
Private Sector Banks	561791	375104	2410141.00
Public Sector Banks	936952	670849	2005620.00
Small Finance Banks	37276	332	88688.00
Grand Total	**1543236**	**1051375**	**18020689.00**

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables .





Calculated Columns & DAX Measures: Implemented DAX formulas

1) Physical_tran = sum('bankwise comlete details'[atm's onsite])+sum('bankwise comlete details'[atm's offsite]).

2) **Digital_tran** = sum('bankwise comlete details'[pos])+sum('bankwise comlete details'[bharath_qr])+sum('bankwise comlete details'[upi_qr]).

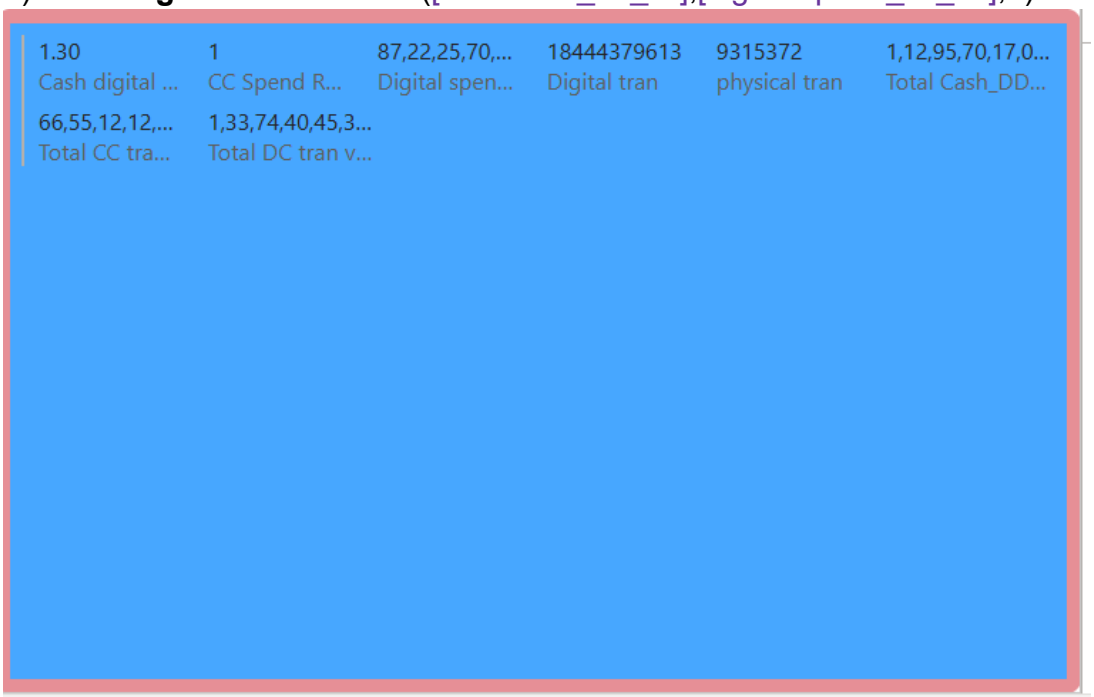
3) **cc spend rank** = rankx(all('bankwise comlete details'[bank_name]), sum('bankwise comlete details'[value of dc at pos]), , desc).

4) **Total CC tran value** = sum('bankwise comlete details'[cc_transaction total value]).

5) **Total DC tran value** = sum('bankwise comlete details'[dc_transaction total value]).

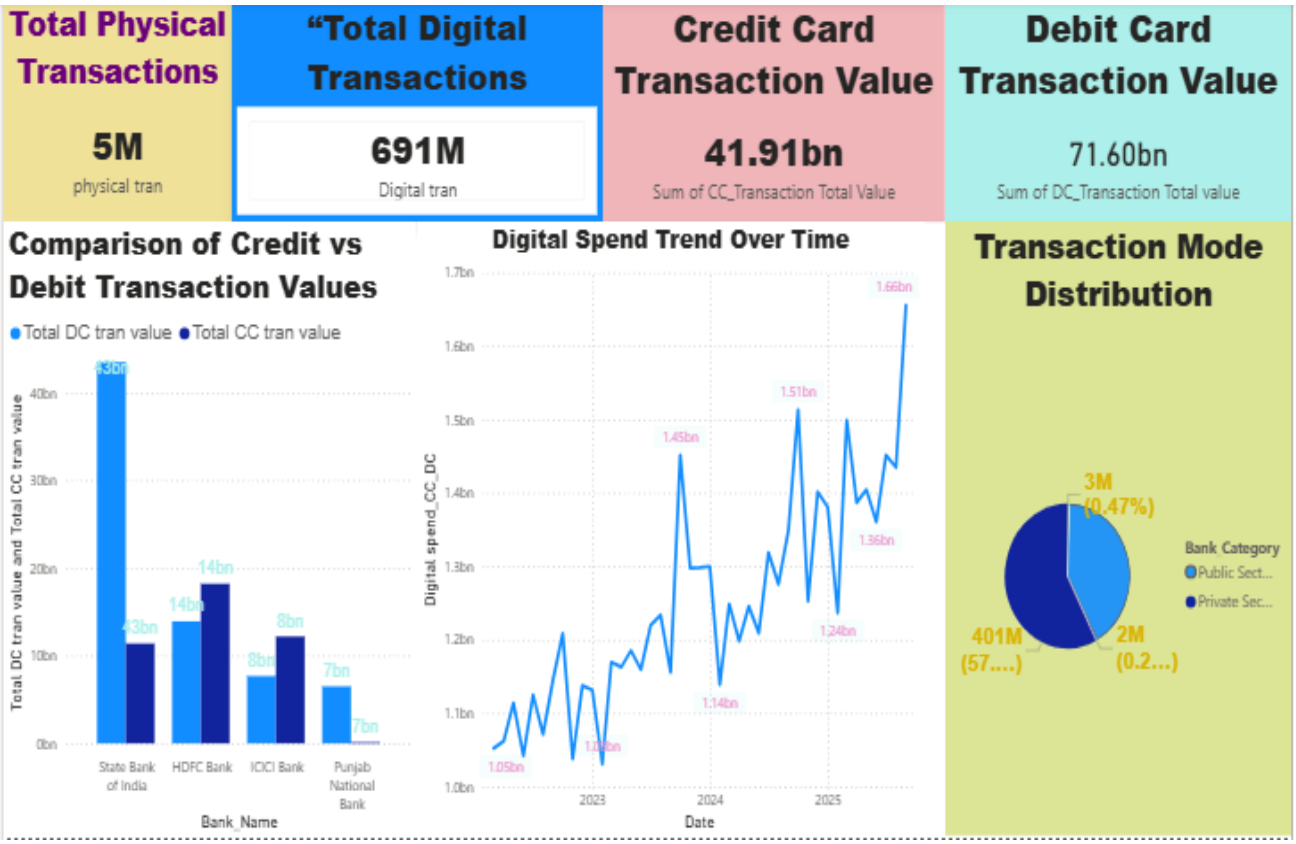
6) **Digital spend_CC_D**C = sum('bankwise comlete details'[cc online transaction value]) +sum('bankwise comlete details'[value of cc at pos])+sum('bankwise comlete details'[value of dc online transaction])+sum('bankw ise comlete details'[value of dc at pos]).

7) **Cash digital ratio** = divide([total cash_dd_cc],[digital spend_cc_dc],0).



8. Analysis and Visualizations (Power BI)

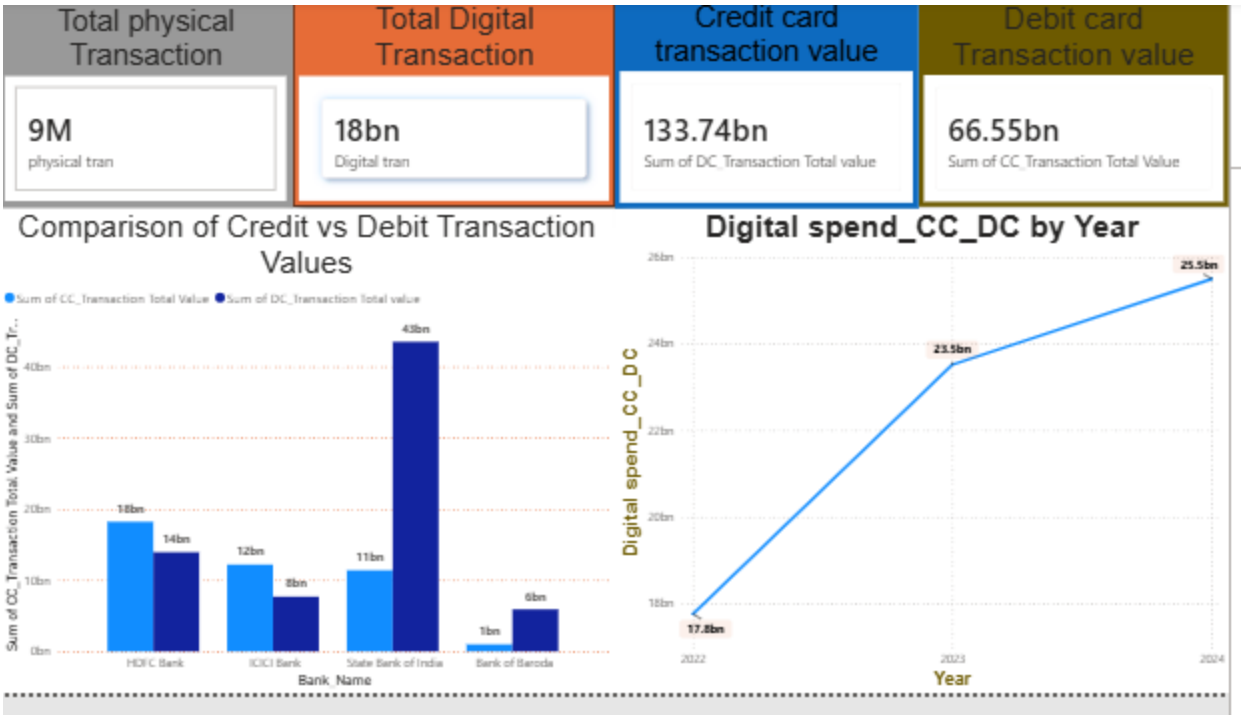
FINANCIAL TRANSACTIONS BREAKDOWN – PHYSICAL, DIGITAL, AND CARD SPEND DASHBOARD

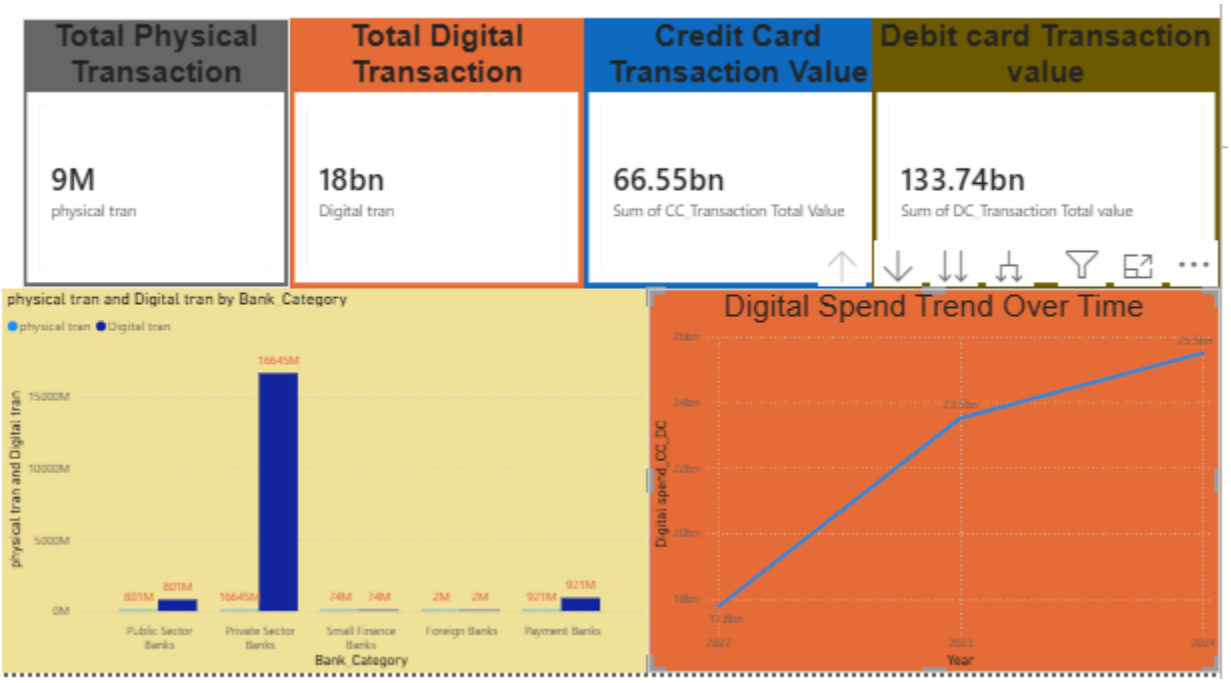


9. Insights

- Physical transactions (ATM withdrawals) are significantly lower compared to digital transactions (POS, QR, UPI).
- Credit card transaction values are higher in certain banks (e.g., HDFC, ICICI) while debit card usage dominates in public sector banks.
- Digital spend shows a steady upward trend over time, indicating customer preference for online and QR-based payments.
- The rise in digital transactions is driven by UPI adoption and QR-based payments, which are faster and more convenient.
- Banks with stronger digital infrastructure (private banks) show higher digital transaction values.
- Physical transactions remain relevant in rural areas due to limited digital penetration and cash dependency.
- Digital transactions will continue to grow, especially UPI and QR payments, as smartphone penetration increases.
- Credit card online spending is expected to rise with e-commerce growth.
- Debit card cash withdrawals may decline gradually as digital alternatives become more accessible
- Banks should invest more in **digital infrastructure** and customer awareness campaigns to accelerate digital adoption.
- Encourage **cashless transactions** by offering rewards/discounts for UPI and card payments.
- Monitor **fraud risks** in digital channels and strengthen security measures.
- Public sector banks should focus on bridging the digital gap by expanding QR/UPI services in semi-urban and rural areas.

(since we have the data of 2025 untill september, 2025 was not considerd in analysis for year growth,for other analysisi its taken)


Total physical Transaction
9M
physical tran
Total Digital Transaction
18bn
Digital tran
Credit card transaction value
133.74bn
Sum of DC_Transaction Total value
Debit card Transaction value
66.55bn
Sum of CC_Transaction Total Value
Comparison of Credit vs Debit Transaction Values
Sum of CC_Transaction Total Value
Sum of DC_Transaction Total value
Sum of CC_Transaction Total Value and Sum of DC_Tr...
40bn
30bn
20bn
10bn
0bn
18bn
14bn
12bn
8bn
11bn
43bn
1bn
6bn
HDFC Bank
ICICI Bank
State Bank of India
Bank of Baroda
Bank_Name
Digital spend_CC_DC by Year
Digital spend_CC_DC
26bn
24bn
22bn
20bn
18bn
17.8bn
23.5bn
25.5bn
2022
2023
2024
Year


Total Physical Transaction
9M
physical tran
Total Digital Transaction
18bn
Digital tran
Credit Card Transaction Value
66.55bn
Sum of CC_Transaction Total Value
Debit card Transaction value
133.74bn
Sum of DC_Transaction Total value
physical tran and Digital tran by Bank_Category
physical tran
Digital tran
physical tran and Digital tran
15000M
10000M
5000M
0M
801M
801M
16645M
16645M
74M
74M
2M
2M
921M
921M
Public Sector Banks
Private Sector Banks
Small Finance Banks
Foreign Banks
Payment Banks
Bank_Category
Digital Spend Trend Over Time
Digital spend_CC_DC
Year

Conclusion

The data reveals a clear shift toward digital payment modes, driven by UPI and QR adoption, especially among private and payment banks. Credit card online spending is poised to grow alongside e-commerce expansion, while debit card cash withdrawals may gradually decline as digital alternatives become more accessible. These trends underscore the need for banks to strengthen digital infrastructure and adapt to evolving consumer preferences.